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*Geology* 1986;14;335-338
doi: 10.1130/0091-7613(1986)14<335:POTUJG>2.0.CO;2

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# Paleomagnetism of the Upper Jurassic Galice Formation, southwestern Oregon: Evidence for differential rotation of the eastern and western Klamath Mountains

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## ABSTRACT

**Paleomagnetic studies of low-grade metatuff within the Upper Jurassic Galice Formation at the type section along the Rogue River, Oregon, indicate stable magnetizations at 13 sites with a mean orientation of D = 245.6°, I = 55.9°, and $\alpha_{95}$ of 5.2°. Site mean directions are well grouped in situ despite the variable orientation of both bedding and cleavage. The magnetizations are therefore secondary and postdate Late Jurassic folding. The most likely times for remagnetization are during low-grade metamorphism of the Late Jurassic Nevadan orogeny (150 Ma) or during intrusion of the nearby Grants Pass pluton (139 Ma). The paleomagnetic pole position calculated from the formation mean direction lies at 40°N, 31°E, far from North American cratonic pole positions aged 150 Ma and younger.**

**Clockwise rotation of the study area by 99° ±10° and tilting or poleward displacement of 0° ±7° are indicated by comparison with a reference pole aged 150 Ma. However, the amount of rotation is not significantly different for other reference poles aged 140 to 80 Ma. Thus, the data indicate that the Galice stratotype and vicinity have rotated approximately 100° since Late Jurassic time, 20° to 40° more than the eastern Klamath Mountains. An unrecognized fault may be required to accommodate the difference in rotation. In addition, restoration of the rotation places Nevadan structural trends in the western Klamaths at a high angle to Nevadan trends in the nearby Sierra Nevada and Blue Mountains of Oregon.**

## INTRODUCTION

The Klamath Mountains (Fig. 1) are formed by one of several Paleozoic-Mesozoic bedrock complexes that underlie Tertiary cover sequences in California and the Pacific northwest region. Similar bedrock complexes occur in the western Sierra Nevada, in the Blue Mountains of Oregon, and in British Columbia (Davis et al., 1978). Paleomagnetic studies indicate that all or parts of the bedrock complexes of the Klamath and Blue mountains have rotated clockwise 50° or more, whereas similar units in the western Sierra Nevada have not rotated (Wilson and Cox, 1980; Achache et al., 1982; Schultz and Levi, 1983; Fagin and Gose, 1983; Mankinen et al., 1984; Hannah and Verosub, 1980; Bogen et al., 1985). Several problems concerning the rotations remain unsolved, including whether they are related to rotations of Tertiary rocks in the Coast Ranges of the Pacific northwest, the timing and cause of the rotations, and whether or not the Klamath Mountains rotated as a single block.

Within the Klamath Mountains, paleomagnetic studies have been performed on two of the three major geologic subprovinces (Fig. 1) recognized by Irwin (1960, 1966, 1981). Results from the eastern Klamath belt (EB, Fig. 1) indicate clockwise rotations of 60° or more (Achache et al., 1982; Mankinen et al., 1984). Pre-Jurassic units may have rotated as much as 100° (Mankinen et al., 1984). In the central part of the range, the Paleozoic and Triassic belt (TrPz, Fig. 1) is intruded by plutons of Late Jurassic to earliest Cretaceous age that have been rotated 87° ±13° clockwise (Schultz, 1983). Paleomagnetic data bearing on the rotation of the third major province, the western belt (WB, Fig. 1), are presented herein.

## REGIONAL RELATIONS

The western belt of the Klamath Mountains is underlain predominantly by basaltic to andesitic volcaniclastic rocks and flows, turbidites, ophiolitic rocks, and lesser silicic volcanics (Wells et al., 1949; Irwin, 1960, 1966, 1981; Garcia, 1979; Harper, 1984, and many others). The Galice Formation is a prominent unit that consists of intercalated volcaniclastic rocks, turbidites, and mudstone (Wells et al., 1949) and contains fossil pelecypods indicative of late Oxfordian to early Kimmeridgian age (Imlay, 1952). The type section is along the Rogue River, Oregon (Wells et al., 1949; Wells and Walker, 1953). The Galice Formation concordantly overlies the Rogue Formation, which consists predominantly of basaltic to andesitic volcaniclastic rocks with lesser mafic to silicic flows. The Rogue Formation is unfossiliferous but is believed to be of approximately mid–Late Jurassic age (Garcia, 1979). The section

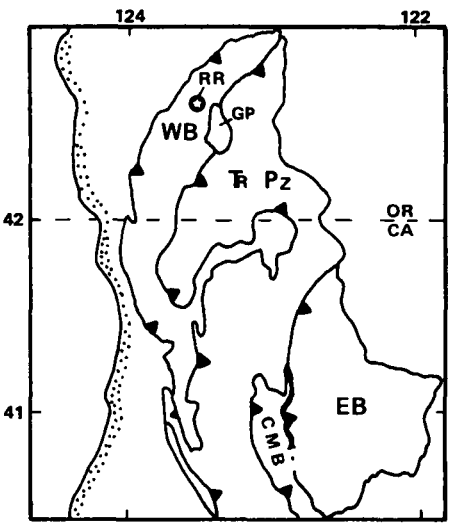



**Figure 1. Geologic provinces of Klamath Mountains region. WB: western belt; TrPz: Paleozoic and Triassic belt; CMB: central metamorphic belt; EB: eastern belt; GP: Grants Pass pluton; RR: Rogue River section. Modified from Irwin (1981).**

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has been deformed and metamorphosed to prehnite-pumpellyite and low greenschist facies during the Nevadan orogeny (Wells et al., 1949; Irwin, 1966; Garcia, 1979).

Rocks of similar age and/or lithology occur to the southeast in the western belt of the Sierra Nevada and include the Mariposa, Logtown Ridge, and Penon Blanco Formations (Taliaferro, 1942; Imlay, 1952; Davis, 1969). The Logtown Ridge and Penon Blanco Formations and the nearby lamprophyric Sonora dike swarm bear a stable secondary magnetization of probable Late Jurassic age (Bogen et al., 1985). Two upper Paleozoic volcaniclastic units in the northern Sierra yield similar results and are also believed to have been remagnetized during the Nevadan orogeny (Hannah and Verosub, 1980). These results indicate that the Sierra Nevada has rotated little or not at all since Late Jurassic time and are in accord with other results from Jurassic and Cretaceous plutons and the Cretaceous Great Valley sequence (Bogen et al., 1985, and references therein). The Galice and Rogue Formations were selected for paleomagnetic study because of their similarities to the Logtown Ridge and Penon Blanco Formations in both lithology and deformational history.

## METHODS AND ANALYSES

Pilot samples of Galice metatuff were collected in the vicinity of the Klamath and Trinity rivers, California (sites 1–5), and the Rogue River, Oregon (sites 7–10). A smaller number of samples of Rogue metatuff were also collected from the Rogue River section (sites 11–13). Results from samples of the Rogue River section, particularly from the Galice Formation, proved most promising, and a second collection was made there (sites 14–23). Sites 6 and 24–28 were collected in other units and are not discussed here. All samples were cored in the field with a gasoline-powered drill and oriented with a magnetic compass.

The direction and intensity of the NRM of each sample was measured with an ScT cryogenic magnetometer, except those samples with intensities greater than 150 mA/m, which were measured with a Schoenstedt spinner magnetometer. Intensities ranged from 0.26 to 320 mA/m, and directions occurred in each octant with no strong preferred orientation.

A suite of pilot samples for demagnetization studies was selected from samples most strongly magnetized. Subsets of this suite were subjected to either alternating fields (AF), thermal demagnetization, or both. High temperatures proved to be most successful in breaking down the magnetization, although both AF demagnetization and thermal demagnetization followed by AF treatment were satisfactory in some cases.

All demagnetization data were plotted on orthogonal vector end-point diagrams to determine the characteristic magnetization direction (Zijderveld, 1967). Characteristic directions were determined from the diagrams by regression analysis. Samples that failed to decay linearly to the origin on such diagrams were rejected unless they yielded stable end-point directions at 10% or less NRM intensity. Sites were rejected if fewer than two samples yielded characteristic directions. On the basis of these criteria, all sites from the Klamath and Trinity river regions (1–5) were rejected, and four sites (7, 10, 12, 13) from the Rogue River region were rejected.

TABLE 1. GALICE FORMATION SITE MEANS AND FORMATION MEAN DIRECTION

| Site | Rs | D | I | k | $\alpha_{95}$ | Str/Dip (o) |
|---|---|---|---|---|---|---|
| 8 | 6/6 | 240.2 | -59.8 | 83 | 7.4 | 118/45W |
| 9 | 2/7 | 233.4 | -55.3 | -- | ---- | 356/49E |
| 11* | 3/6 | 235.5 | -53.7 | 35 | 15.8 | 12/79E |
| 14 | 5/6 | 251.6 | -59.4 | 46 | 11.4 | 118/45W |
| 15 | 3/5 | 256.2 | -51.8 | 74 | 14.4 | " |
| 16 | 4/5 | 235.0 | -61.2 | 43 | 14.1 | " |
| 17 | 5/6 | 258.7 | -60.1 | 27 | 15.0 | " |
| 18 | 4/4 | 237.3 | -67.9 | 35 | 15.8 | " |
| 19 | 5/5 | 240.0 | -61.8 | 70 | 9.2 | " |
| 20 | 3/4 | 253.3 | -36.6 | 24 | 25.8 | 356/49E |
| 21 | 3/6 | 263.7 | -46.0 | 11 | 38.6 | " |
| 22 | 2/6 | 240.3 | -49.4 | -- | ---- | " |
| 23 | 4/5 | 237.6 | -58.4 | 51 | 13.0 | " |
| Formation mean: | | | | | | |
| | 13/23 | 245.6 | -55.9 | 65 | 5.2 | |

Note: Rs = ratio of samples averaged to those collected (or sites, for formation mean); D = paleodeclination; I = paleoinclination; k, $\alpha_{95}$ = statistical parameters of Fisher (1953). Mean site location: lat 42.6°N, long 236.5°E. Paleomagnetic pole location: 40°N, 310°E.
*Site 11 is in the Rouge Formation.

Forty-nine samples from 13 sites yielded characteristic or stable end-point magnetization directions. Sample BKM8-3a (Fig. 2A), for example, has a NRM directed downward toward the west but decayed approximately to the origin between 350 and 560 °C, yielding a characteristic magnetization directed upward to the southwest. Similar results are shown for samples BKM16-3a and 18-3a (Fig. 2B, C). Unblocking temperatures of these samples range from 350 to 565 °C, an indication that magnetite is the probable carrier of the magnetization. Four samples yielded characteristic magnetizations directed downward to the northeast, approximately antiparallel to the southwesterly directions.

Pilot studies indicated that samples with low NRM intensities were best treated at low temperatures or low alternating fields. Sample BKM9-1a (Fig. 2D), for example, decayed toward the origin in alternating fields up to 35 mT. Twelve samples of low initial intensity yielded characteristic magnetizations between 100 and 300 °C.

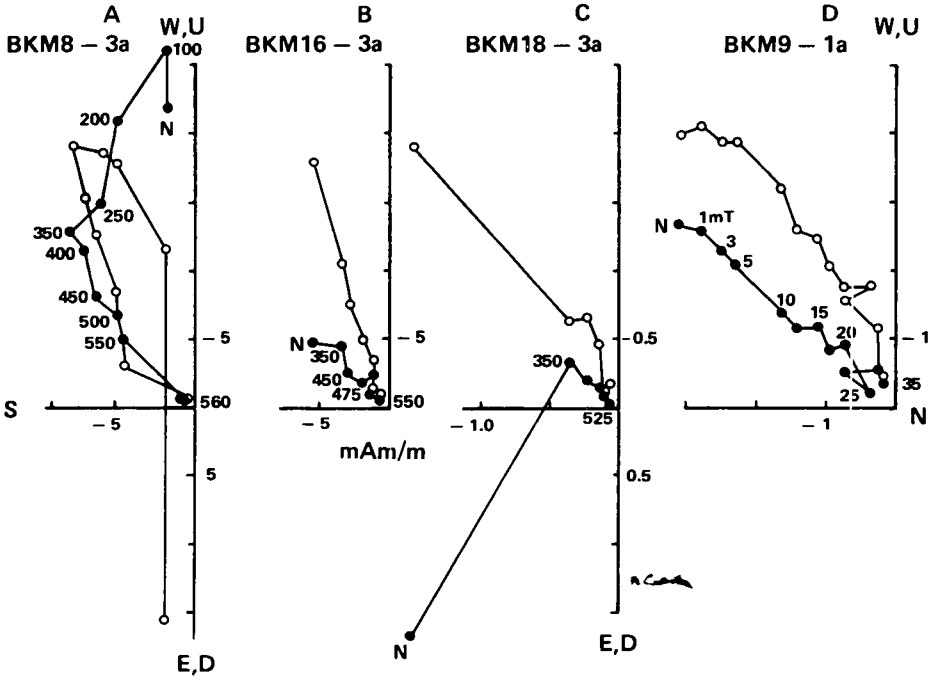



**Figure 2. Orthogonal vector end-point diagrams of stepwise demagnetization data. Temperatures are in degrees Celsius and intensities in mA/m. Solid circles are on horizontal planes; open circles are on vertical planes.**

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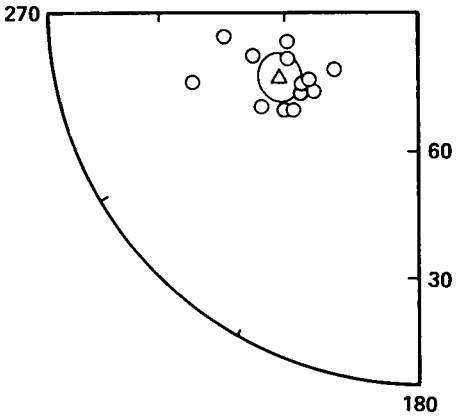



**Figure 3. Galice Formation site (circles) and formation (triangle) mean directions with cone of 95% confidence. Upper-hemisphere, equal-area projection.**

### Formation Mean Direction

The mean magnetization direction determined at each site is listed in Table 1 and shown in Figure 3. All sites yielded mean directions upward to the southwest. The four northeasterly, downward-directed samples are from different sites, and their antiparallel directions were used to obtain the mean directions. The mean of the site mean directions (Table 1; Fig. 3) has an orientation of D = 245.6°, I = 55.9°. The semi-angle of the cone of 95% confidence about the mean and the precision parameter (Fisher, 1953) are 5.2° and 65, respectively. These results indicate little variation in direction between sites and suggest that secular variation has been averaged out in large part. One site in the Rogue Formation (site 11, Table 1) yielded a mean direction not significantly different from sites in the Galice Formation and is included in the Galice Formation mean direction.

### Age of the Magnetizations

Site mean directions are tightly clustered despite the variable orientation of bedding and cleavage through the section (Table 1). Bedding and cleavage are subparallel, and their orientations commonly change in concert (Wells and Walker, 1953), suggesting that a second set of folds is present; a second, penetrative cleavage occurs locally. A conventional fold test disperses the magnetization directions and is negative at the 95% confidence level. This simple test lacks rigor, however, because there are two sets of folds, and the locations of their hinge zones are not well known. It seems highly unlikely that correction for two phases of folding would reduce the scatter of the site means significantly; thus, the magnetizations are almost certainly secondary and probably postdate both sets of folds.

The age of deformation and low-grade metamorphism in the western Klamath Mountains is well established as Late Jurassic or, more precisely, Kimmeridgian to early Tithonian, about 150 Ma (Lanphere et al., 1968; Saleeby et al., 1982). In California, the Galice Formation has yielded K-Ar metamorphic ages of 148 to 153 Ma (Lanphere et al., 1978). The two sets of cleavages in the Rogue River section both contain aligned plates of chlorite (my unpublished data); therefore, it is likely that both sets of folds formed during the Nevadan orogeny because no younger episode of dynamic metamorphism is known in the region. Remagnetization of the section, therefore, can be firmly bracketed only as 150 Ma or younger.

Two thermal events have occurred in the region of the Rogue River that may have been responsible for remagnetization of the section. The older is the Nevadan metamorphism, and the younger is the intrusion of the nearby Grants Pass pluton (Fig. 1) which has a K-Ar age of 139 Ma (Hotz, 1971). The pluton is 7 to 15 km from the sampling sites, however, and seems a less likely cause of remagnetization than Nevadan metamorphism, which is known to have directly affected all sites and to have reset K-Ar ages in northern California about 50 km to the south. Nevadan metamorphism is also the likely cause of widespread remagnetization of similar metavolcanic rocks within the western Sierra Nevada (Hannah and Verosub, 1980; Bogen et al., 1985). Thus, remagnetization during the Nevadan orogeny is likely, but ages of 139 Ma or younger cannot be ruled out.

## DISCUSSION AND CONCLUSIONS

The Galice Formation mean direction yields a paleomagnetic pole located at 40°N, 310°E which is far removed from poles for the North American craton aged 150 Ma or younger given by Harrison and Lindh (1982) or Irving and Irving (1982). This strongly suggests that the Rogue River region has undergone substantial tilting, translation, and/or rotation since Late Jurassic time. Small erosional remnants of nearly flat-lying Albian sedimentary rocks unconformably overlie the western belt and adjacent TrPz at several localities within southwestern Oregon (Ramp and Peterson, 1979, and references therein); thus, substantial tilting of the region during the past 100 Ma can be ruled out, although farther to the east and southeast, the Cretaceous Hornbrook Formation has been tilted 10° to 30° (Nilsen and Barats, 1984).

Comparison of the Galice Formation mean direction with expected directions determined from North American poles aged 150 and 140 Ma shows that the discrepancy is predominantly in the paleodeclination. The range of possible anomalies between the observed and expected paleoinclinations obtained from any reference pole aged 150 Ma or younger given by Harrison and Lindh (1982) or Irving and Irving (1982) is −1° to 15°, and the range of possible anomalies in paleodeclination (for clockwise rotations) is 66° to 99°. Thus, the evidence indicates that the study area has rotated substantially with respect to cratonic North America, regardless of the time of remagnetization.

The time of remagnetization favored here is 150 Ma, as discussed above. North American poles closest to this in age have been determined from the Morrison Formation and Canelo Hills volcanic rocks (Steiner and Helsley, 1975; Kluth et al., 1982) and have a mean location of 64°N, 144°E, with $\alpha_{95}$ of 7.5°. Compared with this pole, the rotation of the Rogue River section at the 95% confidence level is 99° ±10° and the flattening is 0° ±7° calculated by the method of Demarest (1983). Published reference poles aged 150 to 80 Ma yield rotations ranging from 93° to 99°, not significantly different.

The western belt and the TrPz to the east are intruded by the Grants Pass pluton (Fig. 1), which ties the two belts together since the beginning of Cretaceous time. This suggests that the large clockwise rotation of the Galice Formation also affects the TrPz, a hypothesis supported by paleomagnetic data from the pluton that indicate clockwise rotation of 78° ±26° and no significant flattening (Schultz, 1983). (Errors at the 95% confidence level are corrected following Demarest, 1983.) The mean rotation inferred from the Grants Pass pluton and three other Late Jurassic plutons that intrude the TrPz in California and Oregon is 87° ±13° (Schultz, 1983). The magnitude of the rotations of the four plutons and the Galice Formation overlap substantially, which indicates that the western belt and TrPz, joined by the Grants Pass pluton, have rotated together some 90° to 100° clockwise since latest Jurassic or earliest Cretaceous time. It is not certain, however, whether this conclusion applies to the southernmost parts of the belts in California.

Previously published paleomagnetic data have engendered several hypotheses about rotation of the Klamath Mountains and its subprovinces that have been discussed recently by Mankinen et al. (1984). Data presented here suggest an alternative hypothesis, that the northwestern Klamaths have rotated 20° to 40° clockwise more since Late Jurassic time than the southeastern part of the range, which has rotated some 60° to 70° clockwise during the

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same interval (Achache et al., 1982; Mankinen et al., 1984). This difference in rotation may have been accommodated by an unrecognized fault between the TrPz and the eastern belt or, alternatively, the whole range may have rotated as a unit 60° to 70° clockwise accompanied by 20° to 40° clockwise oroclinal bending of the northwestern region with respect to the eastern belt. Additional data from rocks of Jurassic or Early Cretaceous age in the southwestern part of the range are necessary to distinguish between these two hypotheses and to better constrain the timing of the rotations.

The large clockwise rotation of the Klamath Mountains creates problems for the paleogeographic reconstruction of the region, particularly for the remnants of the Nevadan orogen. Nevadan structural trends are north-northwesterly in the western Sierra Nevada to the south and are oriented similarly in the Blue Mountains to the north after correction for 50° to 60° clockwise rotation (e.g., Dickinson and Thayer, 1978; Wilson and Cox, 1980). Correction of the Klamath rotations, however, results in structural trends that are highly oblique to those of the flanking bedrock complexes. This implies that the Nevadan orogen did not form as a linear mountain belt and that the Klamath Mountains did not connect directly to the north with the Blue Mountains of Oregon. More likely, their northward continuation was in the Stikine or Quesnell terranes of British Columbia which were transported northward during Cretaceous and early Tertiary time by right slip on the Tintina–Rocky Mountain trench fault system (Davis et al., 1978; Gabrielse, 1985). It is possible that clockwise rotation of the Klamath and Blue mountains occurred in response to that right slip.

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## ACKNOWLEDGMENTS


Supported in part by the Turner Fund, Department of Geological Sciences, University of Michigan. I thank Valerian Bachtadse and Rob Van der Voo for their reviews and comments and Chris Weiss and Elmar Bruneniecks for assistance in the field.

Manuscript received June 3, 1985
Revised manuscript received December 19, 1985
Manuscript accepted January 7, 1986